MultiPC: Multitask Learning for Joint Pansharpening and Classification From Multispectral and Panchromatic Images

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***Abstract*—Multitask learning (MTL) has broad applicability in remote sensing. Although MS/PAN-based pansharpening and classification tasks are highly complementary at the feature level, they are usually modeled separately. To this end, this article proposes the multitasking parallel architecture of MultiPC to achieve collaborative optimization. To address inconsistency among tasks, MultiPC introduces the mutually enhanced feature fusion (MEFF). For differential features between tasks, it achieves feature alignment through deformable convolution and combines overlapping and cross-attention to enhance feature complementarity between tasks. MEFF simultaneously introduces a joint optimization strategy to improve the synergy and complementarity of tasks. Furthermore, we propose the reflective learning loss function to effectively perceive the learning states across various tasks. It guides the two tasks to adaptively optimize the direction based on the historically learned knowledge, thereby clarifying the learning status of each task and promoting the overall collaborative optimization. Experiments on multiple real remote sensing datasets show that MultiPC significantly outperforms existing single-task methods in both classification and pansharpening tasks. The code is open from: https://github.com/Xidian-AIGroup190726/MultiPC**

***Index Terms*—Classification, contrastive loss, multitask learning, panchromatic and multispectral images, pansharpening.**

I. Introduction

With the advancement of satellite technology, the demand for remote sensing images in applications such as environmental monitoring, agricultural assessment, and land cover classification continues to grow [1], [2]. High-resolution optical satellites provide both panchromatic (PAN) and multispectral (MS) images, offering higher spatial and richer spectral information. Their complementary properties enable more comprehensive feature representation and have been widely used in tasks such as pansharpening and classification [3].

Pansharpening methods can be broadly divided into traditional and deep learning (DL)-based approaches [4], [5], [6]. Traditional methods such as component substitution (CS [7]) are efficient but prone to spectral distortion, while multiresolution analysis (MRA, e.g., wavelet [8], Laplacian pyramid [9]) better preserves spectra but suffers from high complexity and limited adaptability. Model-based methods (e.g., Bayesian [10], matrix factorization [11]) offer theoretical flexibility but rely on strong priors. In contrast, DL methods—such as PNN [12], MSDCNN [13], p2sharp [14], DSPNet [15], GAN-based PSGAN [16], and UCGAN [17]—achieve superior performance but demand large datasets, heavy computation, and still face generalization challenges.

For classification methods, DL has also become mainstream due to its superior performance [18], [19], [20]. ReSC-Net [21] introduces an online hyperspectral data augmentation strategy and a novel spatial-channel attention mechanism to improve feature extraction. SAN, proposed by Liu et al. [22], adaptively selects appropriate patch scales for different objects. FDGNet, developed by Qin et al. [23], integrates frequency disentanglement and data geometry into a domain generalization framework for cross-scene hyperspectral image classification (HSIC), enhancing robustness via adversarial training.

Despite recent progress, most existing methods are developed under a single-task learning (STL) paradigm. As remote sensing applications become increasingly complex, STL suffers from limited representation capacity and optimization inefficiency. Independently trained models often duplicate low-level feature extraction, leading to redundancy, while the complementary information across related tasks remains underexploited.

Multitask learning (MTL) addresses this issue by jointly optimizing multiple related tasks through shared representations, enabling effective knowledge transfer and feature generalization [24]. In remote sensing, MTL has been applied to tasks such as infrared small-target detection and segmentation [25], as well as offset field extraction and 3-D structure reconstruction [26], demonstrating its potential for improving efficiency and generalization.

However, research on jointly optimizing pansharpening and classification remains limited. Multiobjective optimization methods, such as DMOEAD [27], balance objectives explicitly but suffer from poor generalization and limited adaptability. Moreover, most existing MTL approaches adopt a unified encoder [25], ignoring the semantic and granularity

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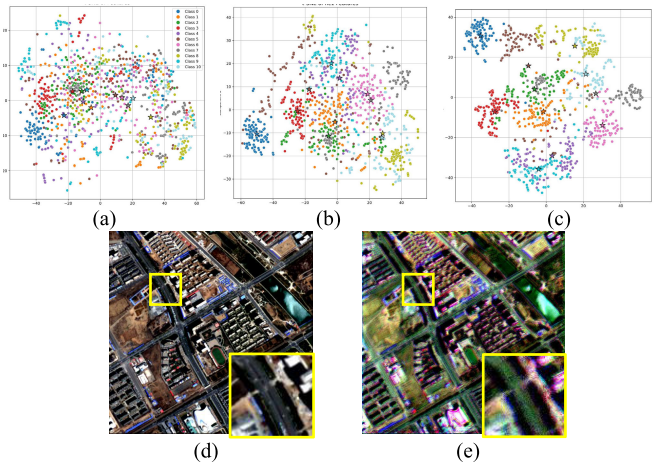


Fig. 1. (a)–(c) correspond, respectively, to P-only (pansharpening single task), C-only (classification single task), and P-C Joint (pansharpening and classification) on Hohhot dataset. In the P-only setting, classes exhibit significant overlap and dispersion due to the sample-dependent spatial details emphasized by pansharpening. The C-only setting yields clearer but still loose clusters, as it focuses on high-level semantics. P-C joint effectively balances both, where semantic guidance reduces intraclass variance and spatial details enhance interclass separability, highlighting task complementarity. As shown in (d) and (e) on Hohhot dataset, P-C joint produces more precise and detailed results than P-only, further confirming this complementarity.

discrepancies between tasks, which restricts effective collaboration. In addition, prior studies primarily rely on LR-HSI inputs, leaving MS/PAN-based joint modeling of pansharpening and classification largely unexplored.

Pansharpening and classification represent typical low-level and high-level tasks, respectively. While pansharpening focuses on restoring high-frequency spatial details, classification emphasizes global semantic representation. Despite their different objectives, the two tasks are highly complementary (see Fig. 1). Semantic supervision from classification can provide structural guidance for pansharpening, whereas the high-frequency details recovered by pansharpening enhance local discriminability for classification. These observations motivate the exploration of a unified modeling framework for collaborative optimization of both tasks.

To summarize, existing joint pansharpening–classification frameworks in MS/PAN scenarios still suffer from two key limitations.

1) Most methods rely on a unified encoder, which overlooks the intrinsic differences in receptive fields and semantic levels between low-level (pansharpening) and high-level (classification) tasks, thereby weakening effective information sharing and complementary modeling.
2) Current multitask optimization strategies struggle to explicitly perceive task-wise learning states and collaboration demands, often degenerating into unstable or single-task-dominated optimization.

To address these issues, we propose MultiPC, an end-to-end concurrent multitask learning framework for joint pansharpening and classification, with the following contributions.

1) We introduce the task feature mutual gain module, which leverages feature alignment and overlapping cross-attention to explicitly model task differences while enhancing cross-task information exchange, enabling more effective complementary collaboration.
2) We design a reflective learning loss that incorporates historical task knowledge to guide current optimization, making task learning states more explicit and ensuring stable, coordinated optimization across tasks.

The remainder of this article is organized as follows. Section II reviews related work, Section III presents the proposed method, Section IV reports experimental results and analysis, and Section V concludes the article.

II. Related Work

A. Pansharping and Classification Methods

In recent years, DL [28], [29], [30], [31], [32] have been widely applied in remote sensing, promoting significant advances in pansharpening (PS) and image classification. For pansharpening, GPPNN [33] introduces a model-based deep network using gradient projection to model PAN-MS relationships, while MDCUN [34] enhances interpretability through a contextual memory mechanism. Panformer [35] applies transformers to PS, leveraging self- and cross-attention for effective feature fusion. GAN-based models such as PSGAN [16] and UCGAN [17] further balance spatial detail and spectral fidelity, with UCGAN [17] enabling unsupervised training via a hybrid cyclic loss. Flow-based and progressive methods, such as PanFlowNet [8] and P2Sharpen [14], improve flexibility and multiscale consistency.

For classification, DSTD [36] employs a dual-stream transformer with difference-aware attention to reduce redundancy and improve accuracy. Ma et al. [37] propose an adaptive hybrid fusion network that combines data- and feature-level strategies, while GCF-Net [38] adopts a patchless global framework with hierarchical and probabilistic fusion. Other studies, such as S2ENet [39] and GLT-Net [40], focus on spatial–spectral enhancement and global–local feature modeling across modalities.

B. Multitask Learning

The rapid emergence of increasingly complex single-task models has led to notable performance gains, yet these approaches are gradually approaching a performance bottleneck. In response, MTL has gained growing attention, as it can leverage complementary information among related tasks to improve performance. For instance, Zhou et al. [41] proposed a multitask framework for joint tumor segmentation and classification in ABUS images, achieving superior performance compared with single-task models. Similarly, Xu et al. [42] introduced RMTL-Net, which segments and classifies tumor regions simultaneously using a region attention module to enhance feature representation.

Beyond medical imaging, MTL has also been explored in remote sensing and infrared vision. Gao et al. [43] jointly learns semantic segmentation and height estimation using cross-task and cross-pixel contrastive losses, achieving state-of-the-art results on Vaihingen and Potsdam datasets.

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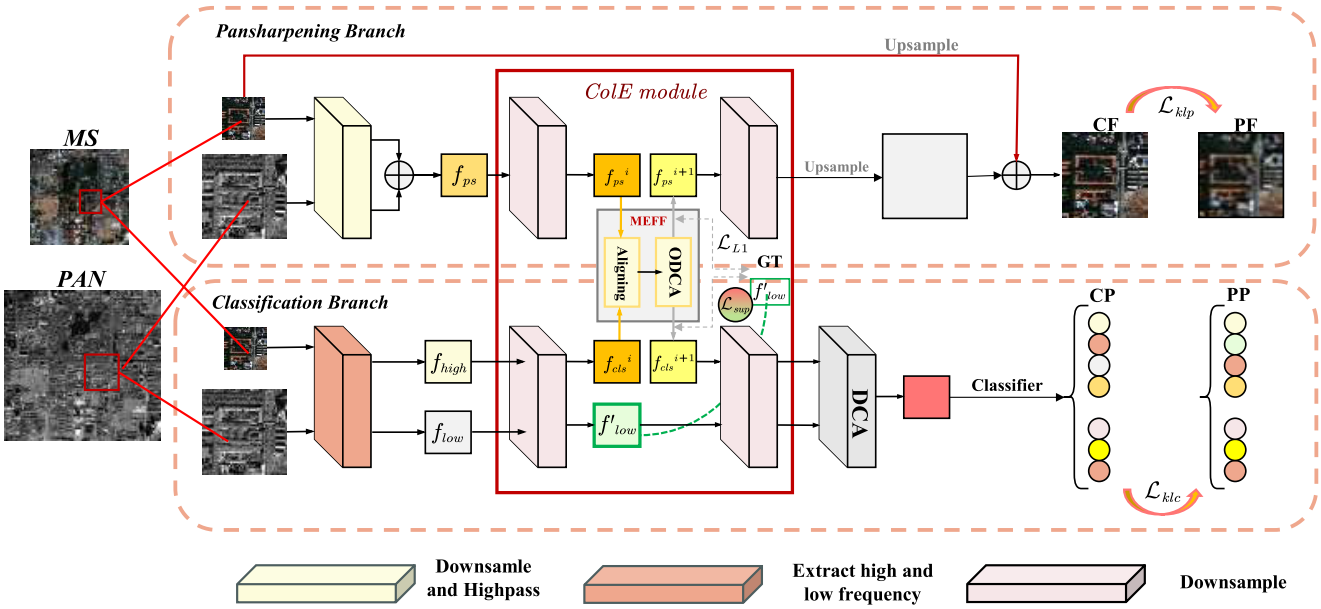


Fig. 2. Architecture of the proposed MultiPC. MultiPC is divided into two branches: the upper branch is the pansharpening branch, and the lower branch is the classification branch. Interaction between the two branches is carried out through MEFF. Pansharpening serves as reflective learning: L_{klp} compares the current figure (CF) to the previous figure (PF); similarly, L_{klc} compares the current predict (CP) to the previous predict (PP).

Chen et al. [25] develop a multitask framework for infrared small target detection and segmentation, which improves detection accuracy, reduces model complexity by nearly half, and speeds up inference compared with single-task models. These examples demonstrate that MTL effectively exploits shared representations and task complementarity, overcoming the limitations of single-task approaches and offering a promising direction for complex vision tasks.

III. Method

A. Architecture

In order to cross the limitations of single-task feature extraction and to build bridges of correlation between modules and tasks, we propose a two-branch parallel architecture, MultiPC. It utilizes MS$\in \mathbb{R}^{W\times H\times 4}$ with PAN$\in \mathbb{R}^{4W\times 4H\times 1}$ as the input. MultiPC consists of three main parts: the pansharpening branch, the categorization branch, and the mutually enhanced feature fusion (MEFF).

1) Pansharpening Branch: The pansharpening task requires enhancing spatial information while reducing color distortion. Based on this, we designed a pansharpening branch dominated by high-frequency information. As shown in Fig. 2, the input is concatenated in the channel dimension after downsampling and high-pass filtering to obtain f_{ps}. The subsequent main processes are as follows:

$$CF = Up\left(MS\right) \oplus Up_n\left(ColE_p^n\left(f_{ps}\right)\right) \tag{1}$$

where $ColE_p^n$ represents the cooperative enhancement strategy performed n times (set n to 2), containing two downsamples and an MEFF module for each branch. The $\oplus$ denotes the splicing of channel dimensions. $Up(\cdot)$ represents the upsampling operation, and CF_t is the Current Feature after training epoch t.

It is worth noting that the pansharpening branch does not learn in isolation; instead, it aligns features with the classification branch at each $ColE_p^n$ using MEFF. It utilizes the more discriminative high-frequency semantics in classification tasks to guide pansharpening enhancement, effectively alleviating color drift.

Furthermore, to steadily improve the quality of image fusion in multitask learning, consider the current epoch t. The current feature representation CF_t learns its spatial structure and detail distribution by referring to the feature result PF_{t-1} of the previous epoch $t-1$

$$\mathcal{L}_{klp} = L_1\left(CF_t, PF_{t-1}\right) \tag{2}$$

where $\mathcal{L}_{klp}$ takes $\mathcal{L}_{L1}$ as the steady-state constraint.

This branch combines high-frequency explicit separation and time series reflection, improving spatial resolution and training stability while ensuring spectral consistency.

2) Classification Branch: The classification task needs to extract more discriminative features, and the input is consistent with the pansharpening task. This branch separates the low-frequency f_{low} and high-frequency f_{high} information using the discrete wavelet transform (DWT), then performs feature extraction and classification. The core process is as follows:

$$f_{cls} = DCA\left(ColE_c^n\left(f_{high}, f_{low}\right)\right) \tag{3}$$

where DCA is dual cross-attention. Specifically, DCA pays attention to both the low-frequency semantic structure and the high-frequency local detail information simultaneously, thereby improving the discrimination ability of features.

The classification branch introduces spatial texture into the pansharpening branch via MEFF at each round of feature extraction, thereby enhancing edge recognition and fine-grained classification. Finally, the classifier outputs the prediction result CP_t of the current epoch.

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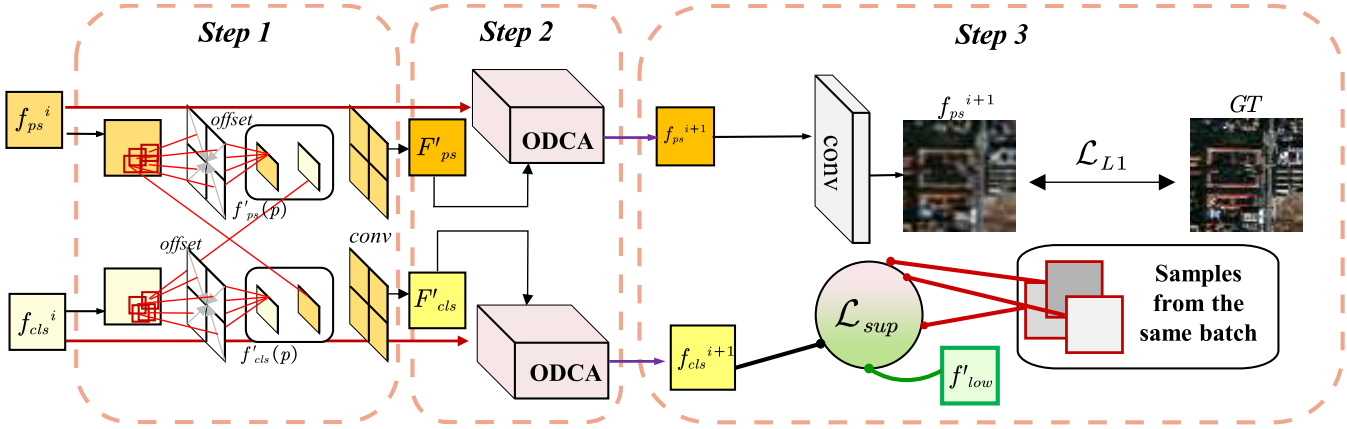


Fig. 3. Structure of the mutually enhanced feature fusion unit. It consists of three steps: 1) feature alignment between two tasks using deformable convolution; 2) ODCA for fine-grained modeling and task interaction, combining overlapping windows as well as dual cross-attention; and 3) MEFF loss, including classification contrast loss $\mathcal{L}_{sup}$ and pansharpening reconstruction loss $\mathcal{L}_{L1}$.

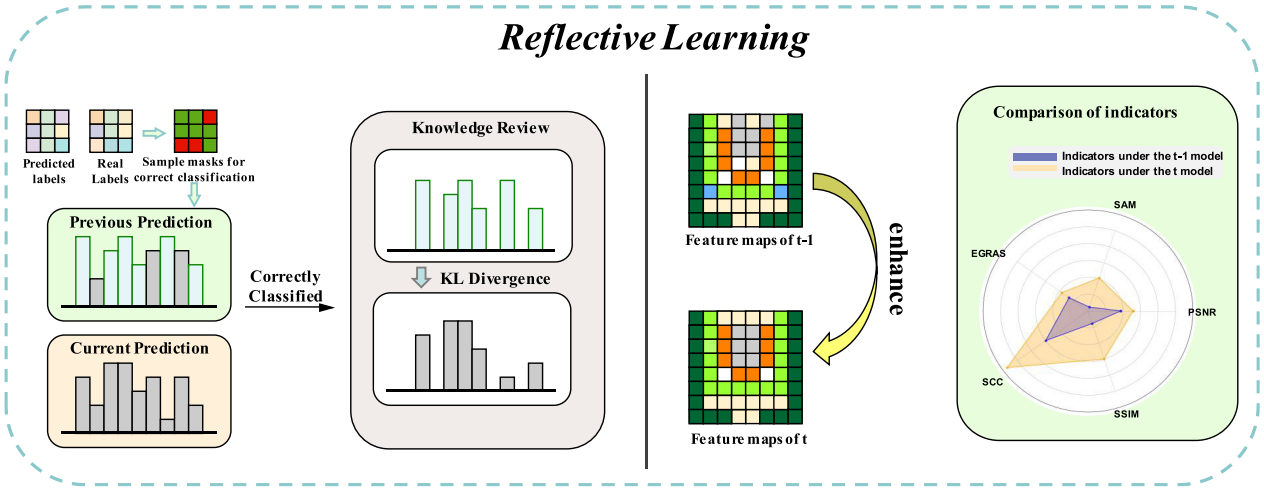


Fig. 4. Reflective learning structure: (left) classification, (right) pansharpening. For classification, a KL scatter is computed by comparing the current-round predictions with the correctly classified samples from the previous round, ensuring that the current-round predictions do not deviate too much from the learned knowledge. For pansharpening, as image quality improves, an L1 loss is computed by comparing the current-round images with those of the previous round.

To improve the training stability and optimize the consistency, the consistency constraint before and after epochs based on the KL divergence is introduced

$$\mathcal{L}_{klc} = KL(CP_t, PP_{t-1}) \tag{4}$$

where PP_{t-1} is the previous prediction for epoch $t-1$. This branch enhances feature discriminability by combining the reflection mechanism with dual cross-attention.

3) Mutually Enhanced Feature Fusion (MEFF): High-level and low-level tasks are naturally complementary in the feature space, but existing methods do not sufficiently explore their cross-task interactions. To address this, we propose the MEFF.

As shown in Fig. 2, MEFF takes the image detail feature ${f_{ps}}^i$ from the pansharpening branch and the semantic feature ${f_{cls}}^i$ from the classification branch, where i denotes the ith MEFF module. It enhances task collaboration through three steps: task-specific feature alignment, window-based cross-attention modeling, and MEFF loss. The MEFF loss includes the semantic reconstruction loss $\mathcal{L}_{L1}$ and the supervised contrastive loss $\mathcal{L}_{sup}$.

This structure builds a bidirectional feedback loop between pansharpening and classification. The formal design is detailed in Section III-B.

B. Mutually Enhanced Feature Fusion Unit

To alleviate the previously mentioned problems, this article proposes the multitask feature mutual gain module MEFF (as shown in Fig. 3). This module adopts a three-stage mechanism to promote effective collaboration and information complementarity among multiple tasks.

1) Aligning: In multitask learning, due to differences in goal orientation and feature perception across tasks, the features extracted by them often exhibit misalignment or spatial offsets. Therefore, we introduced the deformable convolution mechanism. Given the intermediate feature map f_{cls}^i from the classification branch and the intermediate feature map f_{ps}^i from the pansharpening branch as input. This operation can be formalized as

$$f'_{cls}(p) = \sum_{k=1}^{K} w_k \cdot {f_{cls}}^i(p + p_k + \triangle p_k) \tag{5}$$

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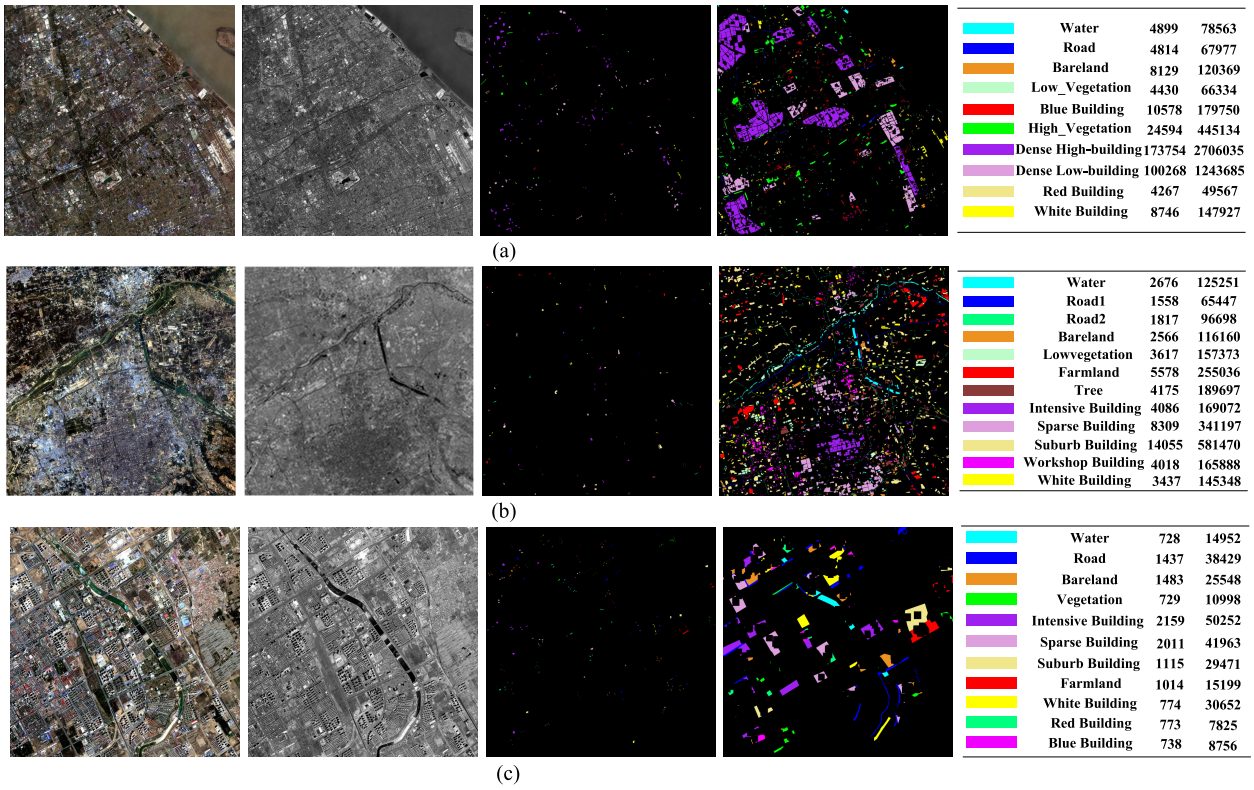


Fig. 5. (From top to bottom) (a) Shanghai dataset, (b) Xi'an dataset, and (c) Hohhot dataset. (From left to right) MS images, PAN images, the labeled training set, the ground truth of the test set, and the distribution of the number of each category (the penultimate column is the number of samples in the training set, and the penultimate column is the number of samples in the test set).

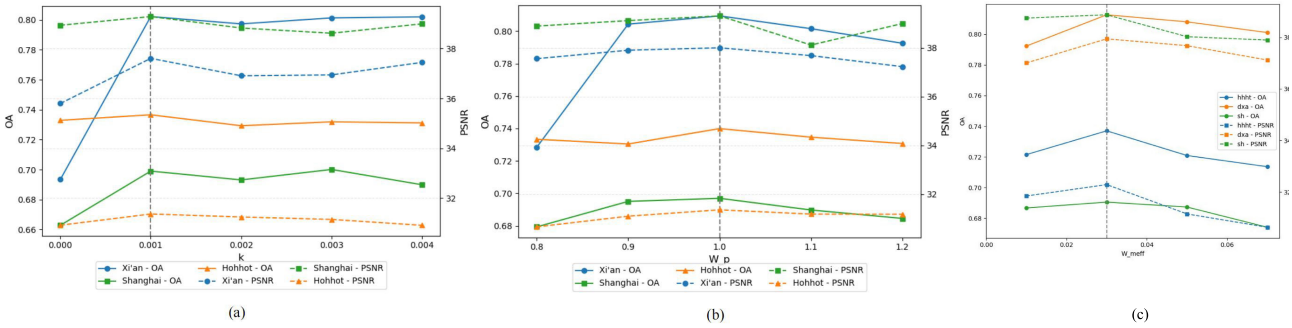


Fig. 6. (a) Results for different values of k for $W_p = 1.0$. k takes values from 0 to 0.004. (b) Results for different values of W_p for $k = 0.001$. The values of W_p range from 0.8 to 1.2. (c) Results for different values of W_{meff} for $k = 0.001$. The values of W_{meff} range from 0.01 to 0.07.

$$f'_{ps}(p) = \sum_{k=1}^{K} w_k \cdot f_{ps}{}^{i}(p + p_k + \triangle p_k) \tag{6}$$

where p denotes the current spatial location, p_k is the relative position of the kth sampling point in the conventional convolution kernel, Δp_k is the shared offset dynamically generated through an offset prediction module, w_k is the weight of the convolution kernel, and K represents the total number of sampling points in the convolution kernel. Perform consistent spatial sampling on the features of the two tasks to achieve cross-task awareness at the aligned spatial positions.

This mechanism can alleviate the spatial dislocation problem between tasks. The core lies in the fact that the two branches share the same Δp_k and can adaptively learn based on image content and task differences. This is crucial for cross-task feature fusion, especially in remote sensing images with local texture variations, scale differences, or boundary offsets.

After deformable convolution aligns the features, direct use cannot ensure the whole interaction. Therefore, convolution is employed to adaptively fuse features from both tasks, promoting effective collaboration. We fuse the original features of each task with the aligned features from the other task to obtain the enhanced feature representation

$$F\prime_{ps} = Conv\left(\left[f_{ps}{}^{i}, f'_{cls}(p)\right]\right) \tag{7}$$

$$F\prime_{cls} = Conv\left(\left[f_{cls}{}^{i}, f'_{ps}(p)\right]\right). \tag{8}$$

By integrating the two, it reflects enhancing the characteristics of the current task itself with the features from the other task. The enhanced features $F\prime_{ps}, F\prime_{cls}$ are used as the input for the subsequent modules.

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TABLE I
QUANTITATIVE EXPERIMENTAL CLASSIFICATION RESULTS OF DIFFERENT METHODS ON SHANGHAI DATASET (BOLD: BEST RESULTS AND UNDERLINED: SECOND BEST RESULTS)

Methods	NNCNet	MFT	AM3Net	CALC	GCFNet	CRHFF	Ours
c1	73.74	73.37	28.93	75.28	71.34	**81.87**	76.28
c2	42.24	17.57	5.38	40.33	41.16	44.86	**45.99**
c3	81.86	73.71	6.82	84.08	83.16	80.49	**85.21**
c4	80.23	61.44	24.62	75.55	69.92	79.53	**81.74**
c5	75.75	70.15	52.60	74.01	83.82	**84.95**	84.46
c6	80.85	79.09	82.36	59.42	82.26	**84.19**	82.46
c7	70.32	73.83	**94.09**	59.14	64.22	69.55	74.67
c8	47.70	39.75	10.84	**64.34**	45.87	50.47	52.48
c9	36.40	**53.42**	35.81	29.47	10.20	52.47	28.96
c10	96.15	92.77	83.05	85.90	93.49	97.06	**97.93**
OA(%)	66.42	65.29	65.29	62.24	62.66	67.60	**69.05**
AA(%)	68.52	63.51	42.45	64.75	64.54	70.54	**71.02**
Kappa(%)	52.60	46.37	38.23	44.76	43.35	52.00	**56.61**
Times(ms)	1723.11	**308.97**	1712.7	562.98	2727.74	2477.0	989.07
Para(M)	2.23	0.21	2.78	**0.11**	3.90	4.91	35.07

This cross-task alignment mechanism promotes mutual benefit among branches. It alleviates spatial-feature misalignment, enhances feature interaction and sharing, and further provides strong support for task collaborative optimization.

2) Overlapping Dual Cross-Attention: When facing the differentiation between high-level and low-level tasks, the existing methods still affect the further exploitation of the potential of multitask collaboration. Based on this, we introduced overlapping dual cross-attention (ODCA) based on window partitioning.

Specifically, take the Q, K, and V after the aligned feature mapping as the input. Q, K, and V are divided into overlapping windows of size $M_0 \times M_0$

$$M_0 = (1 + \rho) \times M \tag{9}$$

where ρ controls the window overlap ratio, and the consistency of the spatial coverage of the feature maps is ensured by symmetric zero-padding $(\rho M)/2$.

Compared to STL, multitask learning typically involves tasks that rely on features at different scales. This disparity between large and small receptive fields can lead to representation mismatch, which is particularly amplified at window boundaries. To address this, we partition the projection into overlapping local windows. Shared windows mitigate the abrupt transitions caused by segmentation to some extent, enabling feature extraction near boundaries based on shared local context. This reduces spatial misalignment of cross-task features, thereby stabilizing subsequent multitask interactions.

After completing the window division of Q, K, V, perform the following operations based on the dual cross-attention strategy

$$A \leftarrow B : Attn\,(Q_A, K_B, V_B) \tag{10}$$

$$B \leftarrow A : Attn\,(Q_B, K_A, V_A) \tag{11}$$

where both directions are applied: A attends B, and B attends A. This dual interaction enables the two tasks (e.g., pansharpening and classification) to guide each other and mutually refine their representations.

Finally, the features after alignment and fusion, together with the original information, are used as the input of DCA and then concatenated along the channel to obtain the output $f_{ps}^{i+1}, f_{cls}^{i+1}$.

Overall, this design promotes adaptive interaction among tasks while retaining their respective semantic characteristics.

3) MEFF Loss: To facilitate intertask feature interaction and guide MEFF to focus on standardized information, we propose MEFF Loss. It includes the supervised contrast loss $\mathcal{L}_{sup}$ for classification tasks and the L1 loss $\mathcal{L}_{L1}$ for semantic reconstruction after pansharpening.

Classification is a high-level task. To enhance the discriminative ability of features, supervised contrastive loss can be utilized to construct positive and negative sample pairs

$$\mathcal{L}_{sup} = \frac{-\sum_{i=1}^{N} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp\left(sim\left(F_i, F_p\right)/\tau\right)}{\sum_{a \in A(i)} \exp(sim(F_i, F_a)/\tau)}}{N} \tag{12}$$

where N represents the number of samples in the current batch, and τ is the temperature coefficient, which is used to adjust the sensitivity of the similarity distribution and is set to 0.1 here. F_i represents the f_{low} extracted from the i th sample in the classification branch, which is used to construct the similarity relationship in the feature space. Based on the label information, for each sample i, we construct the positive sample set $P(i)$, that is, the index set of samples belonging to the same category as it. In contrast, the remaining samples form the candidate set $A(i)$, which is used to participate in the comparison calculation.

Compared with high-frequency features, low-frequency features are less disturbed by local textures. It can better reflect a stable semantic structure, thereby prompting the network to focus on category-discriminative features consistent with the classification target.

We use the semantic enhancement feature ${f_{ps}}^{i+1}$ after fusion with the classification task for low-level pansharpening. the reconstruction is supervised via an L1 loss against the ground truth GT

$$\mathcal{L}_{L1} = \frac{1}{N} \sum_{i=1}^{N} \left\| {f_{ps}}^{i+1} - GT \right\|_1. \tag{13}$$

This prompts the pansharpening branch to obtain global structural guidance from cross-task fusion, enhancing the overall rationality of the reconstruction results.

Overall, the MEFF module constructs a multitask feature fusion mechanism with deep collaboration through three steps. This mechanism alleviates the feature offset between tasks caused by differences and enhances the semantic expression of classification and the ability to reconstruct the details of pansharpening.

C. Reflective Learning

Traditional methods, such as EMA and KD, either struggle to distinguish between correct and incorrect samples in multitasks or find it difficult to maintain multitask stability. To address this, we propose reflective learning, which reviews historical performance to enhance recognition and maintain training stability.

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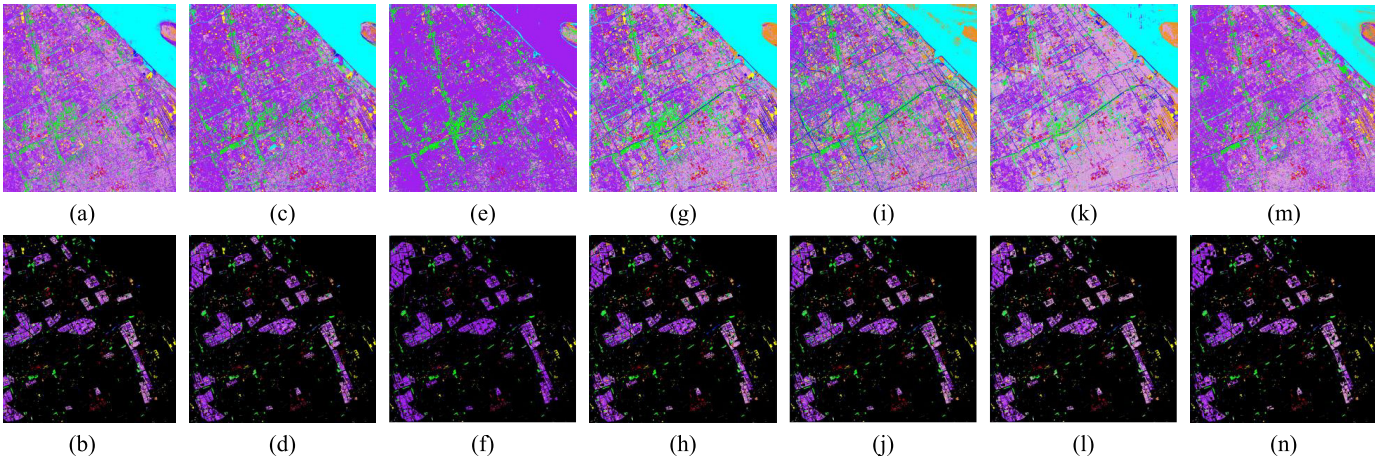


Fig. 7. Classification maps of different methods on Shanghai dataset. (First line) Classification maps for the overall image. (Second line) Classification maps on the test set. (a) and (b) NNCNet. (c) and (d) MFT. (e) and (f) AM3Net. (g) and (h) CALC. (i) and (j) GCFNet. (k) and (l) CRHFF. (m) and (n) Ours.

During multitask training, the gradient directions across different tasks may diverge. This can lead to shared parameters being repeatedly pulled toward each other's optimal directions, thereby causing the forgetting of existing knowledge. Therefore, we take the predicted distributions of the current model and the historical model as input, and the operation is as follows:

$$KL\left(p_{i,t-1} \parallel p_{i,t}\right) = \sum_{i=1}^{n} p_{i,t-1}\left(x_i; \Theta_{t-1}\right) \log \frac{p_{i,t-1}\left(x_i; \Theta_{t-1}\right)}{p_{i,t}\left(x_i; \Theta_t\right)} \tag{14}$$

where $p_{i,t-1}(x_i; \Theta_{t-1})$ represents the prediction probability distribution of x_i at train step t using the model with the parameter Θ_{t-1}. Minimizing the KL divergence between the predicted distributions allows the model to retain historical knowledge while learning new samples. This enables the two tasks to be optimized more smoothly in tandem.

To prevent erroneous semantics from becoming entrenched over time, we transfer and extract feature information solely from correctly classified samples. As shown in Fig. 4, reflective learning uses predictive labels and accurate labels to generate sample masks for correct classification. A set of correctly categorized instances (CCIs) is then defined based on this definition

$$\mathbb{A}_{CCI} = \{x_i \in \mathbb{A} | arg\max\left(p_i\left(x_i; \Theta\right)\right) == y_i\} \tag{15}$$

where y_i denotes the ground-truth label of instance x_i, and $\mathbb{A}_{CCI}$ contains all correctly classified instances. Based on the correct prediction of the historical model, we integrate L_{klc} using the CCI set as

$$\mathcal{L}_{klc} = \frac{\sum_{x_i \in \mathbb{A}^{t-1}{}_{CCI}} KL\left(p_{i,t-1} || p_{i,t}\right)}{\left|\mathbb{A}^{t-1}{}_{CCI}\right|}. \tag{16}$$

This preserves reliable semantics and demonstrates that reflective learning is not merely a repetition of training steps.

Because in the pansharpening task, the image quality steadily improves. Therefore, we only compare the pansharpening results of the current model with those of the historical model to ensure the stability and progressive optimization of the generated images (as shown in Fig. 4). The calculation formula is

$$\mathcal{L}_{klp} = \frac{\sum_{x_i \in \mathbb{A}} \left\| out\left(x_i; \Theta_t\right) - out\left(x_i; \Theta_{t-1}\right) \right\|}{|\mathbb{A}|} \tag{17}$$

where $\mathbb{A}$ contains the set of all samples. The reflective learning in the pansharpening aims to constrain the consistency between the output $out(x_i; \Theta_t)$ of the current train step and $out(x_i; \Theta_{t-1})$ of the previous train step through the L1 loss.

Therefore, the reflective loss $\mathcal{L}_{rl}$ is defined as follows:

$$\mathcal{L}_{rl} = \lambda_c \cdot \mathcal{L}_{klc} + \lambda_p \cdot \mathcal{L}_{klp}. \tag{18}$$

Since the classification task is unique, each sample corresponds to only one correct label. Therefore, λ_c is constant to ensure the model continuously focuses on the semantic consistency of historically correctly classified samples.

In the early training stage, the historical output of pansharpening is limited. As the training progresses, its guiding role in optimization gradually increases. Therefore, λ_p is a dynamic weight that increases linearly with the number of

TABLE II

QUANTITATIVE EXPERIMENTAL CLASSIFICATION RESULTS OF DIFFERENT METHODS ON XI'AN DATASET (BOLD: BEST RESULTS AND UNDERLINED: SECOND BEST RESULTS)

Methods	NNCNet	MFT	AM3Net	CALC	GCFnet	CRHFF	Ours
c1	93.15	97.35	95.61	97.59	95.89	95.89	**97.98**
c2	**34.16**	16.82	1.72	12.71	17.46	18.35	22.97
c3	**62.14**	50.69	11.61	42.79	55.38	62.01	59.65
c4	69.01	64.58	67.74	39.02	66.46	66.45	**74.72**
c5	55.10	46.68	18.81	**74.46**	36.06	55.51	45.79
c6	68.41	51.69	**76.35**	65.03	64.31	69.74	73.32
c7	77.92	91.78	37.73	80.48	89.12	84.84	**92.36**
c8	74.62	62.85	72.44	70.91	72.65	77.59	**80.58**
c9	77.48	75.73	80.85	**82.01**	76.40	77.18	77.38
c10	96.81	91.90	**99.81**	96.93	97.81	95.51	98.72
c11	94.23	94.52	75.48	**96.74**	90.71	95.39	93.90
c12	76.02	72.87	45.00	74.14	66.36	75.33	**76.25**
OA(%)	79.24	74.65	69.58	78.24	76.86	79.42	**81.27**
AA(%)	73.25	68.12	56.93	69.40	69.05	72.81	**74.47**
Kappa(%)	76.24	71.08	64.18	75.11	73.48	76.44	**78.52**
Times(ms)	756.16	**140.23**	1041.57	199.83	2474.43	1159.44	427.03
Para(M)	2.23	0.21	2.78	**0.11**	3.90	4.91	35.07

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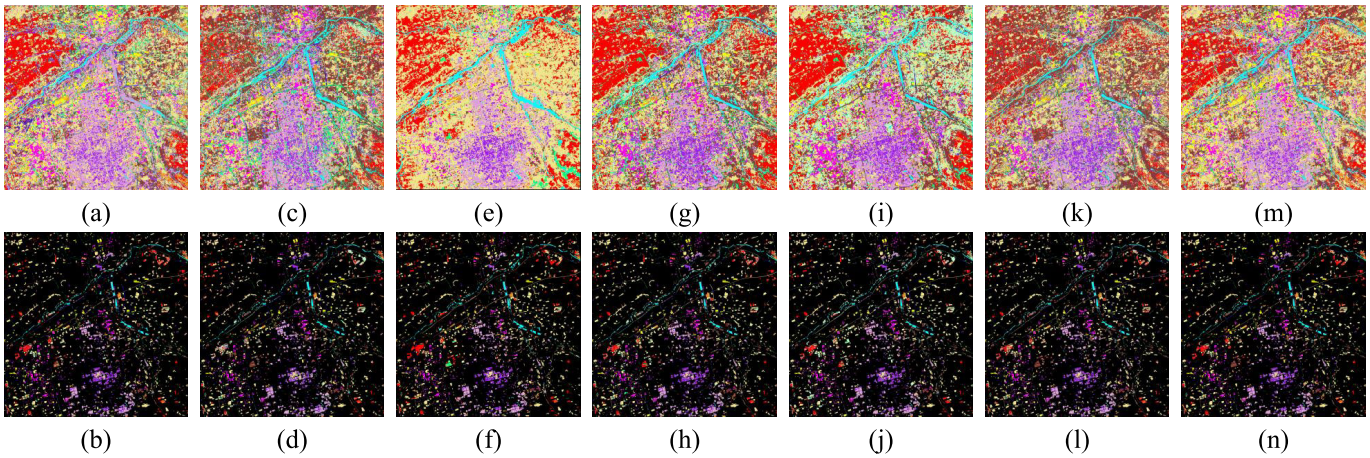


Fig. 8. Classification maps of different methods on Xi'an dataset. (First line) Classification maps for the overall image. (Second line) Classification maps on the test set. (a) and (b) NNCNet. (c) and (d) MFT. (e) and (f) AM3Net. (g) and (h) CRHFF. (i) and (j) CALC. (k) and (l) GCFNet. (m) and (n) Ours.

training rounds to adapt to the optimization requirements at different stages.

That is,

$$\lambda_c = \text{const}, \lambda_p = k \cdot epoch. \tag{19}$$

The overall loss function is

$$\mathcal{L}_{total} = W_c \cdot \mathcal{L}_c + W_p \cdot \mathcal{L}_p + W_{rl} \cdot \mathcal{L}_{rl} + W_{meff} \cdot \mathcal{L}_{meff} \tag{20}$$

where $\mathcal{L}_c$ denotes the cross-entropy, $\mathcal{L}_p$ denotes L1 loss, and W_{rl} and W_c are fixed to 1.

Overall, reflective learning dynamically constrains the current model's training direction by incorporating knowledge from historical models. Compared with traditional methods for task loss weights, our method shows stronger adaptability and robustness during stable training and for guiding multitask collaboration.

IV. Experiments

A. Dataset Description

1) Shanghai Dataset: The Shanghai dataset (as shown in Fig. 5) consists of ten labeled categories, i.e., waters, roads, bare ground, vegetation, sparse buildings, dense buildings, suburban buildings, farmland, white buildings, and red buildings. The spatial resolution of the MS images is 0.8 m (size $6950 \times 7300 \times 4$), while the spatial resolution of the PAN images is 3.2 m (size $27\,620 \times 29\,200 \times 1$). The dataset covers a dense urban area with a wide variety of building materials and vegetation distribution, and is therefore well-suited for evaluating model performance in complex urban environments.

2) Xi'an Dataset: As shown in Fig. 5, Xi'an dataset includes 12 annotation categories such as water, road 1, bare ground, road 2, suburban buildings, dense buildings, white buildings, sparse buildings, trees, low vegetation, farmland, and plant buildings. The MS image has a spatial resolution of 0.8 m (size $4541 \times 4548 \times 4$), while the PAN image has a resolution of 3.2 m (size $18\,164 \times 18\,192 \times 1$). The dataset captures mixed urban and rural areas with rich land cover diversity and different building densities, providing a valuable benchmark for validating the generalisability and robustness of the model.

3) Hohhot Dataset: As shown in Fig. 5, Hohhot dataset contains 11 labeled categories, including waters, roads, bare ground, low vegetation, blue buildings, dense high buildings, dense low buildings, red buildings, white buildings, high vegetation, and farmland. The multispectral (MS) image has a spatial resolution of 3.2 m and a size of $2001 \times 2101 \times 4$, whereas the pansharpening (PAN) image has a spatial resolution of 0.8 m and a size of $8004 \times 8404 \times 1$.

B. Experimental Setups

We used qualitative and quantitative metrics to evaluate the performance of our multitask learning approach. For classification, we adopted overall accuracy (OA), average accuracy (AA), and the Kappa coefficient to assess both global and class-wise performance. For pansharpening, we used SCC, SAM, ERGAS, and PSNR to evaluate both spatial detail fidelity and spectral consistency.

Training samples were generated by randomly cropping the original image into patches centered on target pixels. Each MS patch was of size $16 \times 16 \times 4$, and each corresponding PAN patch was $64 \times 64 \times 1$. The dataset had no overlap between training and test sets and was free of fine-tuning artifacts, as illustrated in Fig. 5.

The model was trained using the Adam optimizer with a fixed learning rate of 0.0001. The loss weights for cross-entropy and L1 loss were both set to 1. We trained for 20 epochs with a batch size of 128. To ensure a fair comparison, all competing methods used their optimal hyperparameters. Each experiment was repeated five times, and the average result was reported.

All experiments were implemented in PyTorch and conducted on a workstation running Ubuntu 20.04 with an NVIDIA GeForce RTX 3090 GPU (24 GB).

C. Hyperparameters Analysis and Selection

In this section, we will use the method of controlling variables to compare and analyze the values of W_p and k.

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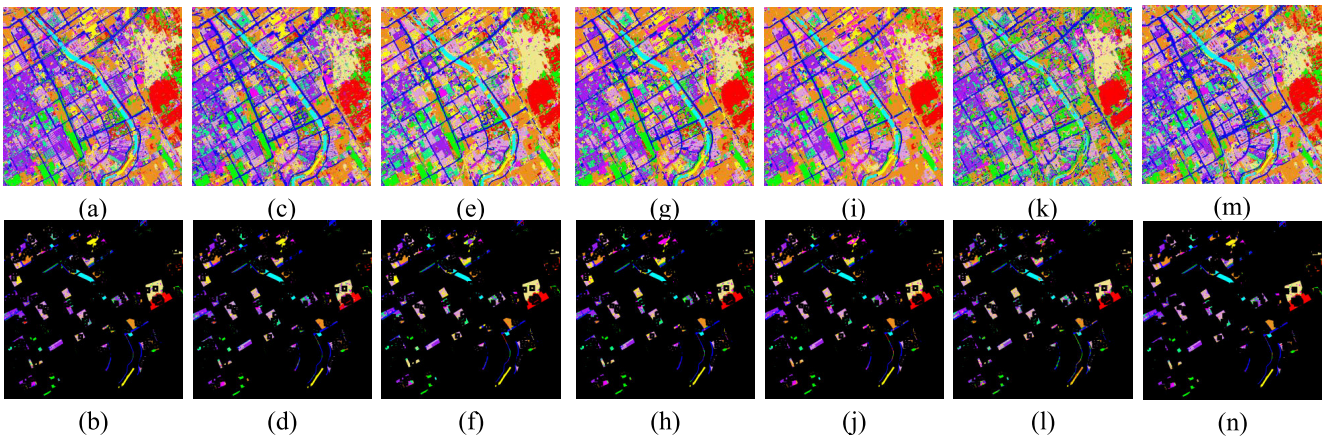


Fig. 9. Classification maps of different methods on Hohhot dataset. (First line) Classification maps for the overall image. (Second line) Classification maps on the test set. (a) and (b) NNCNet. (c) and (d) MFT. (e) and (f) AM3Net. (g) and (h) CRHFF. (i) and (j) CALC. (k) and (l) GCFNet. (m) and (n) Ours.

TABLE III

QUANTITATIVE EXPERIMENTAL CLASSIFICATION RESULTS OF DIFFERENT METHODS ON HOHHOT DATASET (BOLD: BEST RESULTS AND UNDERLINED: SECOND BEST RESULTS)

Methods	NNCNet	MFT	AM3Net	CALC	GCFnet	CRHFF	Ours
c1	87.55	96.18	92.02	**96.91**	86.70	95.80	93.66
c2	75.40	76.46	76.39	70.45	62.20	**81.48**	78.77
c3	84.76	81.99	84.44	**95.02**	68.84	88.48	89.89
c4	85.24	69.03	84.38	82.52	76.07	**87.85**	84.53
c5	**67.63**	54.23	56.88	65.52	58.99	59.57	64.21
c6	60.41	60.83	65.17	**70.69**	61.91	60.56	63.47
c7	73.94	67.83	**86.66**	67.15	70.10	69.98	77.68
c8	91.94	91.36	92.59	92.84	89.51	93.33	**93.41**
c9	52.93	58.79	55.63	50.88	54.46	55.16	**60.51**
c10	57.38	**82.46**	74.00	67.87	32.37	78.50	63.78
c11	59.86	75.04	67.77	72.91	32.03	**77.11**	67.53
OA(%)	70.85	69.33	72.31	72.50	63.77	72.21	**73.70**
AA(%)	72.46	74.02	75.91	75.70	63.01	75.91	**76.13**
Kappa(%)	67.00	65.47	68.95	69.02	58.99	68.95	**70.24**
Times(ms)	83.42	**15.62**	81.62	25.25	265.81	150.86	49.32
Para(M)	2.23	0.21	2.78	**0.11**	3.90	4.91	35.07

1) Slope k of λ_p in (18): First, taking $k = 0.002$ as the median, we varied it from 0 to 0.004 with a step size of 0.001, and the corresponding results are shown in Fig. 6. The best performance across all datasets is achieved when $k = 0.001$. When $k = 0$, the learning weight of $\mathcal{L}_{klp}$ becomes constant, failing to adaptively focus on increasing detail, resulting in a noticeable drop in OA and PSNR. On the other hand, when k is too large, the pansharpening loss dominates in later epochs, which leads to the model neglecting classification and weakens the overall performance.

2) W_p and W_{Meff} in (19): Since the two tasks utilize complementary features and can promote each other, we fix W_c to 1 and only dynamically regulate W_p to find a balanced value that can accommodate both tasks. Therefore, only the tuning result of W_p is shown as a representative in this article. Taking $W_p = 1.0$ as the baseline, we vary it within [0.8, 1.2] at a step of 0.1. As shown in Fig. 6, $W_p = 1.0$ yields the best classification performance on all datasets. Although not optimal for PSNR on Xi'an dataset, the difference is marginal. In contrast, $W_p = 0.8$ or 1.2 leads to clear drops in both OA and PSNR, indicating that excessive or insufficient weighting disrupts task balance and weakens cross-task synergy. Taking $W_{meff} = 0.01$ as the baseline, we adjusted it in the range of [0.01, 0.07] in steps of 0.02. It is found that the best results are achieved when $W_{meff} = 0.03$ as shown in Fig. 6.

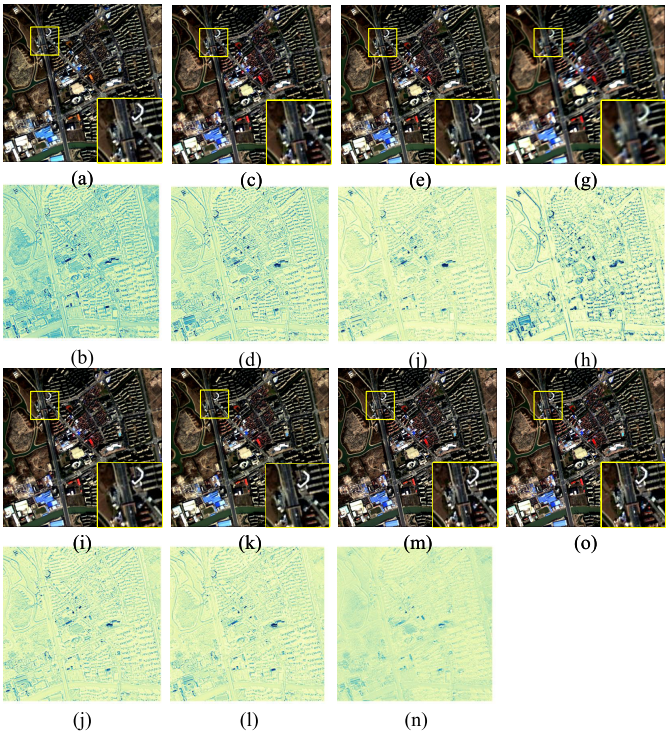


Fig. 10. Visual comparison of the pansharpening results of Shanghai dataset. (Above) Visualization results. (Below) Corresponding RMSE maps. (a) and (b) GPPNN. (c) and (d) MSDDN. (e) and (f) PANFormer. (g) and (h) PANnet. (i) and (j) PSMD. (k) and (l) DCPNet. (m) and (n) Ours. (o) Ground truth.

D. Comparison With Other Classification Methods

1) Experimental Result on Shanghai Dataset: The quantitative and qualitative results of all classification methods on Shanghai dataset are shown in Table I and Fig. 7, respectively. We provide a detailed analysis below.

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TABLE IV
QUANTITATIVE EXPERIMENTAL PANSHARPENING RESULTS ON ALL DATASETS (BOLD: BEST RESULTS AND UNDERLINED: SECOND BEST RESULTS)

Models	Shanghai					Xi'an					Hohhot					Para(M)
	PSNR↑	SAM↓	SCC↑	ERGAS↓	T(ms)	PSNR	SAM	SCC	ERGAS	T	PSNR	SAM	SCC	ERGAS	T	
GPPNN	35.2618	1.7280	0.9129	1.6079	63.74	33.3096	2.4139	0.7472	1.8092	27.99	30.3747	2.7679	0.8910	2.6718	3.30	0.12
MSDDN	34.4127	1.8280	0.8934	1.8127	**30.97**	33.5582	2.2671	0.7570	1.8020	**15.44**	30.5476	2.2037	0.8945	2.1548	**2.19**	**0.02**
PANformer	35.4762	1.5644	0.9156	1.5812	1813.44	34.0024	2.1302	0.7747	1.6905	807.39	30.2466	2.5241	0.8930	2.2358	95.85	1.52
PanNet	29.6325	2.4268	0.7098	3.2193	354.60	31.6186	2.4043	0.6547	2.2851	165.27	26.1686	2.8044	0.7523	3.7367	18.65	0.08
PSMD	35.6202	1.5696	0.9175	1.5625	477.43	34.1795	2.0556	0.7825	1.6692	199.67	30.9584	2.0974	0.9010	2.1353	22.49	0.90
DCPNet	36.1118	1.4765	0.9265	1.4567	3617.75	33.7949	2.1371	0.7774	1.7119	1375.65	31.4667	2.2576	0.9222	2.0018	153.47	1.97
Ours	**38.8756**	**1.2067**	**0.9639**	**1.1238**	989.07	**37.9420**	**1.5257**	**0.9102**	**1.0850**	427.03	**32.2891**	**2.0257**	**0.9282**	**1.8756**	49.32	35.07

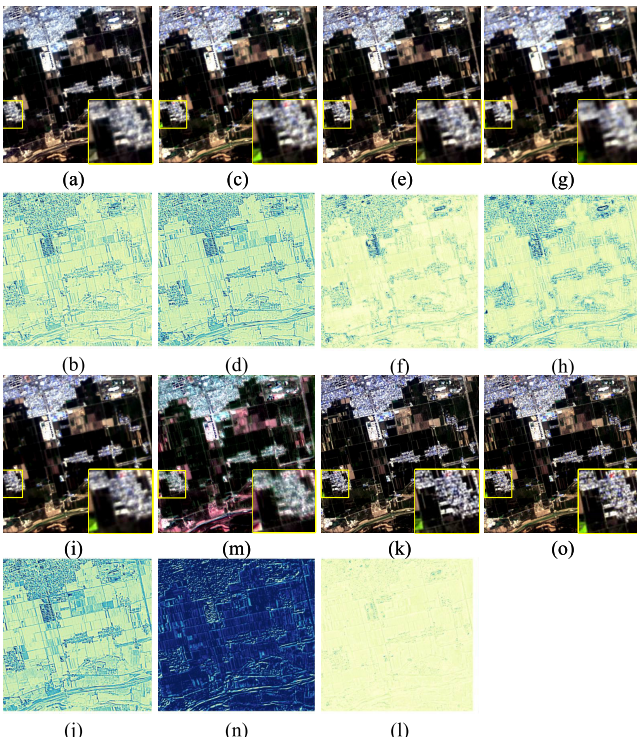


Fig. 11. Visual comparison of the pansharpening results of Xi'an dataset. (Above) Visualization results. (Below) Corresponding RMSE maps. (a) and (b) GPPNN. (c) and (d) MSDDN. (e) and (f) PANFormer. (g) and (h) PANnet. (i) and (j) PSMD. (k) and (l) DCPNet. (m) and (n) Ours. (o) Ground truth.

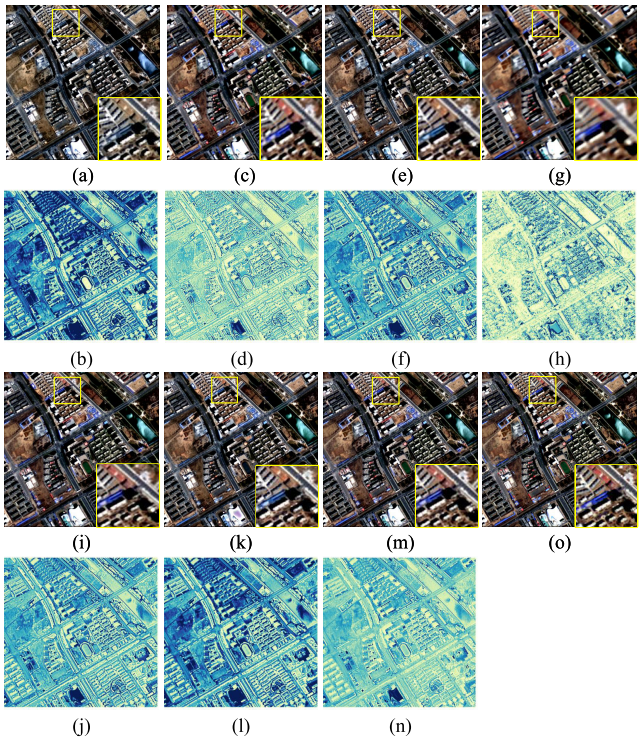


Fig. 12. Visual comparison of the pansharpening results of Hohhot dataset. (Above) Visualization results. (Below) Corresponding RMSE maps. (a) and (b) GPPNN. (c) and (d) MSDDN. (e) and (f) PANFormer. (g) and (h) PANnet. (i) and (j) PSMD. (k) and (l) DCPNet. (m) and (n) ours. (o) Ground truth.

NNCNet leverages unlabeled data to learn semantic relationships between neighboring regions, resulting in strong performance on dense, structured classes such as c2 (road) and c4 (Low_Vegetation). However, its limited sensitivity to abrupt feature changes results in poor performance in complex areas such as c8.

MEF, built on the mCrossPA module, incorporates multimodal inputs to improve spectral-based classification and performs best on c9 (red building). However, its lack of emphasis on high-frequency features reduces its accuracy on c2 and c8.

GCFNet adopts a global patchless strategy that enhances contextual modeling, yielding high accuracy across large, homogeneous regions such as c1 (water) and c3 (bareland). However, due to neglecting local structure, it underperforms in complex categories such as c6 (High_Vegetation) and c7 (dense high-building).

CALC applies coupled adversarial learning to fuse high- and low-level features, showing strong results on uniform regions (e.g., c1, c3). However, insufficient attention to fine-grained variations limits its effectiveness on c6 and c7.

AM3Net introduces involution and spectral-spatial mutual guidance to improve feature interactions. It excels in areas with complex spatial variation, like c6 and c7. However, it shows weakness in planar classes such as c2 (road) and c3 (bareland), likely due to limited perception of flat structures.

Although the model contains 35.07-M parameters, about 60% are allocated to the backbone. It achieves an inference time of 989.07 ms, significantly faster than GCFnet and CRHFF, while providing a 4%–6% accuracy gain over the lightweight MFT at a modest increase in runtime, demonstrating a balanced efficiency–accuracy tradeoff.

2) Experimental Result on Xi'an Dataset: The quantitative and qualitative results of the Classification of our method on

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TABLE V
ABLATION EXPERIMENTS IN THE MEFF SECTION

Align	ODCA	$\mathcal{L}_{meff}$	OA(%)	AA(%)	Kappa(%)	PSNR↑	SAM↓	SCC↑	EGRAS↓
✓			66.58	69.05	48.48	35.0055	1.4662	0.9443	1.5835
		✓	68.08	70.19	50.80	35.3724	1.6860	0.9357	1.4888
✓	✓		68.32	70.82	52.31	37.0321	1.3339	0.9548	1.4120
✓		✓	68.03	70.15	52.72	37.7156	1.2892	0.9583	1.3541
✓	✓	✓	**69.05**	**71.02**	**56.61**	**38.8756**	**1.2067**	**0.9639**	**1.1238**

TABLE VI
ABLATION EXPERIMENTS IN EACH METHODS

MEFF	RL	PA	OA(%)	AA(%)	Kappa(%)	PSNR↑	SAM↓	SCC↑	EGRAS↓
	✓		66.59	69.13	48.51	35.0055	1.9364	0.9128	1.7565
✓		✓	68.54	68.20	51.14	35.2328	1.6292	0.9079	1.7082
	✓	✓	67.52	69.86	51.61	34.5927	1.8159	0.8967	1.8784
✓	✓	✓	**69.05**	**71.02**	**56.61**	**38.8756**	**1.2067**	**0.9639**	**1.1238**

Xi'an dataset are shown in Table II and Fig. 8. After extracting the complementary features of pansharping, our method is applied to c1 (water), c4 (bareland), c7 (tree), c8 (intensive building), and c12 (white building), all of which achieved excellent results. c5 (low vegetation), c9 (sparse building), and c11 (workshop building) did not achieve the highest scores but were not significantly different from the overall suboptimal performance of NNCNet. Despite having more parameters (35.07 M), its inference time (427.03 ms) remains much lower than GCFnet (2474.43 ms) and CRHFF (1159.44 ms), showing high efficiency and accuracy. Compared with the lightweight MFT (0.21 M, 140.23 ms), it achieves notably better accuracy with only a modest increase in runtime.

3) Experimental Result on Hohhot Dataset: The quantitative and qualitative results of the Classification of our method on Hohhot dataset are shown in Table III and Fig. 9. Due to the bi-directional complementarity of feature extraction, our method achieves promising results on c2 (road), c3 (bareland), c4 (vegetation), c7 (suburb building), c8 (farmland), and c9 (white building) with good results. The spectrum may be less sensitive to specific colors, resulting in the model not performing well at c10 (red building) and c11 (blue building). Despite having more parameters (35.07 M), its inference time (427.03 ms) is much shorter than GCFnet (2474.43 ms) and CRHFF (1159.44 ms), demonstrating high efficiency. Compared with the lightweight MFT (0.21 M, 140.23 ms), it achieves higher accuracy with only a slight runtime increase.

E. Comparison With Other Pansharpening Methods

We selected six representative pansharpening methods (GPPNN, MSDDN, PANformer, PanNet, PSMD, and DCP-Net) for systematic comparison with the proposed method. The quantitative results and qualitative comparisons of the methods on the three datasets are shown in Table IV and Figs. 10–12. The experimental results show that our method achieves the optimal performance on all evaluation metrics and is firmly in the first place.

The second and fourth columns in Fig. 10 show the MSE visualization diagrams between each method and GT. The lighter the color, the closer it is to GT. Our method significantly outperforms other methods in reconstructing the contour structure closer to the actual scene. The zoomed-in-view area in Fig. 11 further demonstrates the advantage of the method in preserving the contour details, especially in the complex residential building area, where our method can more accurately reproduce the texture and structural details and also performs better in the road and lake areas.

In addition, Fig. 12 shows the visual results on Hohhot dataset, where our method is the most stable regarding building silhouette reproduction and color reproduction and slightly outperforms the other comparative methods, despite the more incredible reconstruction difficulty in this area. Our method shows good generalization ability and robustness in multiple scenarios.

F. Ablation Study

This section validates the effectiveness of the MEFF module, reflective learning, and multitasking parallel learning. Ablation results are shown in Tables V and VI, where ODCA and MEFF loss correspond to the second and third components of MEFF, respectively. The visualization results are shown in Fig. 13.

1) Effectiveness of the Meff: Table V presents the ablation results on Shanghai dataset. Introducing the $\mathcal{L}_{meff}$ loss or the ODCA module individually leads to notable improvements in both classification and fusion metrics, indicating their effectiveness in enhancing spectral–spatial representation and cross-modal interaction. When all components are combined, the proposed model achieves the best overall performance, with OA, PSNR, and Kappa increasing significantly while SAM and ERGAS decrease, demonstrating that each module contributes complementarily to both accuracy and reconstruction quality.

2) Effectiveness of the Parallel Architecture: By comparing the second and third rows of Table VI, it can be seen that the parallel architecture shows a modest improvement in classification accuracy, and although pansharpening has decreased, the change is not significant. This might be because only the parallel architecture was used for training without using MEFF for information interaction, and the model performance did not change much.

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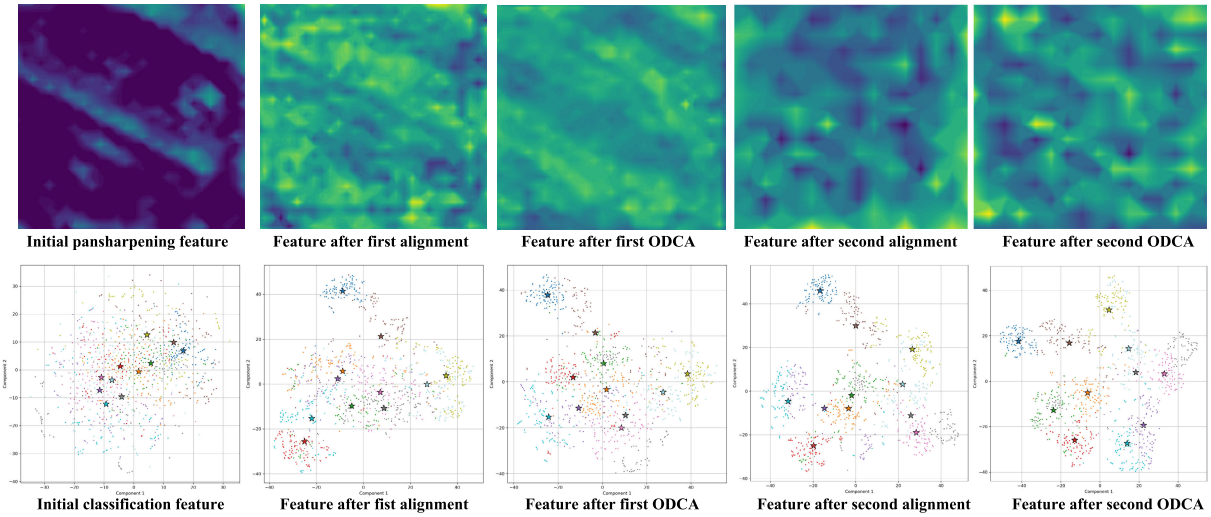


Fig. 13. First column presents the initial feature distributions, while the second and third columns illustrate features after the first MEFF refinement (steps 1 and 2), and the fourth and fifth columns depict features after the second MEFF refinement. Stars represent the cluster centroids for each class.

3) Visualization: As shown in Fig. 13, MEFF strengthens texture features and improves semantic separability. Features within the same class are more compact, and interclass boundaries are more precise in t-SNE space. Contour maps also reflect better edge recovery. This confirms the mutual reinforcement between classification and pansharpening through MEFF interaction.

These results validate the design of each module in the proposed method. The best performance is achieved when all components work together, highlighting the importance of collaborative optimization.

V. Conclusion

To address the performance saturation of low-level and high-level tasks in remote sensing, this article proposes a parallel multitask framework, MultiPC, targeting pansharpening and classification. MultiPC enhances cross-task information sharing and collaborative learning by introducing feature alignment and overlapping cross-attention. A reflective loss is further designed to prevent optimization bias toward a single task, improving training stability. Extensive experiments have verified the effectiveness of this multitask framework in classification and pansharpening tasks, and its performance is superior to that of a single-task. However, as the architecture is tailored to pansharpening and classification, its generalization to other tasks remains limited. Moreover, the number of parameters is substantial. In the future, we plan to develop a lighter weight version by employing more efficient backbone architectures or implementing cross-task parameter sharing.

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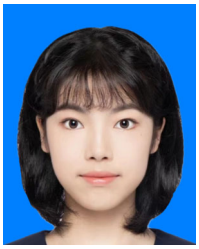
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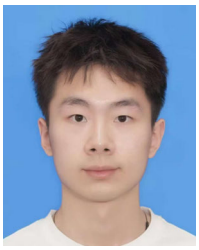
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